

Nitin S Katagihallimath

AI Engineer | AI Agents | LLM Applications
nitinsk0920@gmail.com +91 8073743139
linkedin github portfolio

Summary

AI enthusiast and freelance developer building intelligent applications using LangGraph, LangChain, RAG, and FastAPI. Focused on healthcare AI, AI agents, LLM-powered chatbots, and multimodal deep learning systems.

Technical Skills

- **Programming**: Python (OOP, modular design), Git & GitHub.
- **AI & ML**:Deep Learning,LLMs
- **AI Agents & Workflows**: LangGraph, LangChain, AI Agents, RAG, n8n
- **LLM APIs & Inference**: Groq API, Google AI Studio (Gemini), HuggingFace Transformers, Ollama
- **Backend, Data & Libraries**: FastAPI, Supabase, PyTorch, Pandas, NumPy, OpenCV, FAISS
- **Deployment & UI**: Render, Docker

Projects

AI-assisted Oral Lesion Classification & Clinical Decision Support System *[2026]*

- Curated a **12,000-image** oral lesion dataset; fine-tuned DenseNet169 for 4-class classification at **94.5% accuracy** with Grad-CAM explainability.
- Added Grad-CAM explainability and a LangGraph + FastAPI workflow with a React UI for image upload, risk scoring, and clinical report generation.
- *Stack: Python · PyTorch · DenseNet169 · LangGraph · FastAPI · React · Render*

CoDe-DuINo – AI Arduino Development Assistant *[2026]*

- Built an **LLM-powered AI agent** that autonomously generates, compiles, and uploads Arduino code via tool calling.
- Integrated voice input, pin mapping, and real-time serial logs; cut hardware setup time **from hours to minutes**.
- *Stack: Python · LangGraph · FastAPI · Groq API · arduino-cli · Render*

CLD Sarcopenia Healthcare Chatbot *[2026]*

- Built a **multilingual AI chatbot** (4 languages) for CLD and sarcopenia patient education with assessments and PDF reports.
- Implemented **RAG with FAISS**, Gemini-powered Q&A, JWT authentication, and Supabase report management.
- *Stack: React · FastAPI · LangGraph · Gemini · FAISS · Supabase · Render*

Education

KLE Technological University, Hubballi – B.E. in Electrical and Electronics Engineering *[June 2026]*

- CGPA 8.10

Prism PU Science College, Dharwad – Pre-University (PCMB) : 89.16% *[Apr 2022]*

Shanti Sadan High School, Dharwad – SSLC : 89.92% *[Jul 2020]*

Certifications

- **NLP Specialization – DeepLearning.AI (Coursera)**Link *[Oct 2025]*
- **AI Research Internship – CAIR Lab (Centre for AI Research)** Link *[May 2026]*
- **FastAPI for Machine Learning** Link *[JULY 2026]*

Achievements

- **1st place – Nivista 3.0 Project Exhibition** for Solar-Assisted Backup Propulsion System for EVs [cert]
- **Research paper accepted at ICMLAS 2025** – Text-to-Image GAN (TextControlGAN) [cert]
- **Patent filing in progress** – Smart syrup bottle cap design (healthcare domain)

Languages & Interests

- **Languages**: English · Kannada · Hindi
- **Interests**: Sketching · Table Tennis · Craft & Art · Sci-fi movies · Cooking